CS-471: Machine Learning

Week 2

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Training, Validation and Testing Sets
- Regression Model Performance Evaluation
- Binary Linear Classification

Recap of Previous Lecture

- How do we choose step size in gradient descent?
- How do we solve a linear regression problem?

Training, Validation and Testing

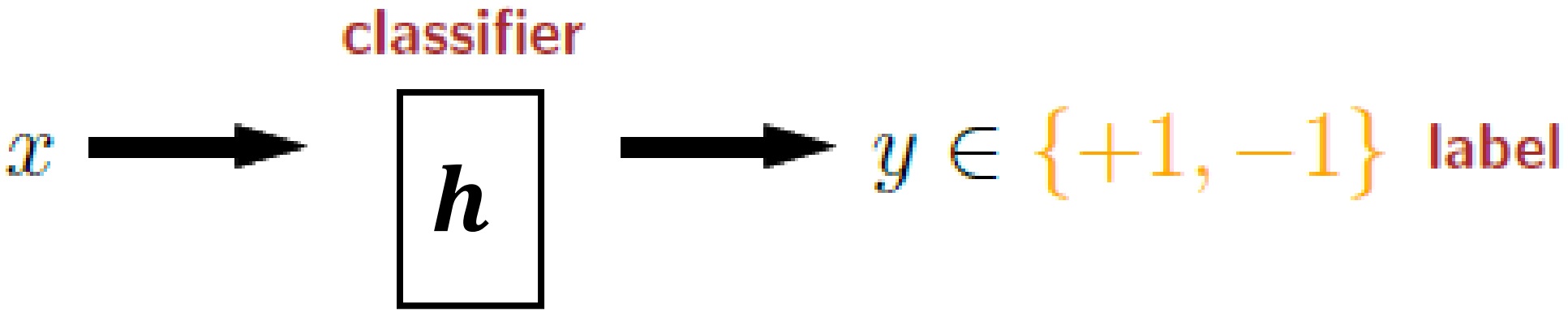
- Dataset is usually split in to training set, validation set and testing set
- Training set is used to train your model and estimate its parameters.
- Validation set is used to validate the performance of your model and tune the hyperparameters.
- Testing set is used to check the accuracy of your final model.
- We need our model to perform well on unseen data.

Regression Model Performance Evaluation

- Mean Squared Error (MSE): Average squared difference between predicted and actual values.
- Root Mean Squared Error (RMSE): Square Root of MSE.
- Mean Absolute Error (MAE): Average absolute difference between predicted and actual values.
- R-Squared (R2): Proportion of variance in target variable explained by model. Ranges between 0 and 1. Higher means better performance.

$$R^2 = 1 - \frac{SS_{residual}}{SS_{total}}$$

Binary Classification Model



- Spam email detection
- Fraud detection in credit card transactions
- In multiclass classification we have more than two labels

Binary Linear Classification

training data

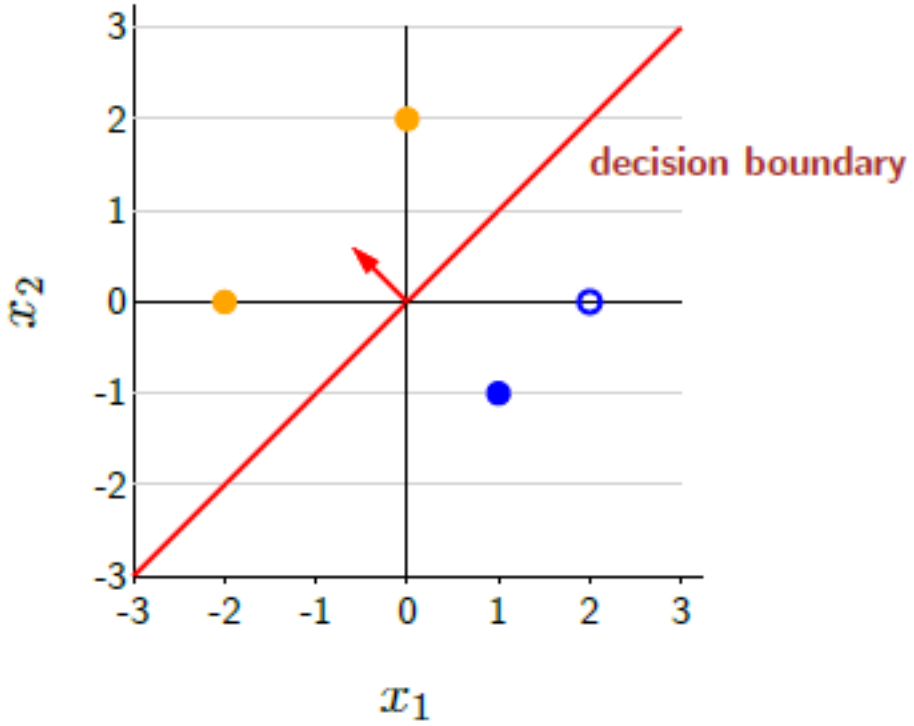
	x_1	x_2	y
example	0	2	1
example	-2	0	1
example	1	-1	-1

learning algorithm

[2, 0] input

h classifier

-1 label



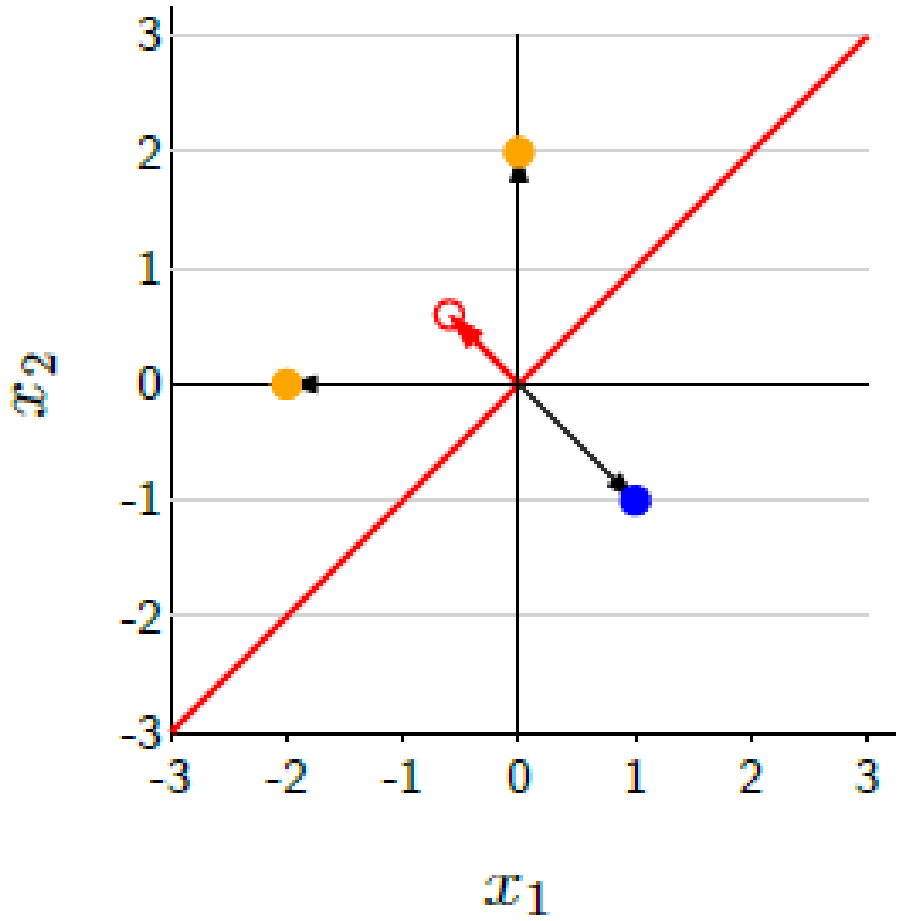
- Hypothesis Class
- Loss Function
- Optimization Algorithm

Example Binary Linear Classifier

- Hypothesis class: $h(\boldsymbol{x}) = sign(\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x}))$ Linear Classifier

Where $sign(z) = \begin{cases} +1 & if \;\; z \geq 0 \\ -1 & if \;\; z < 0 \end{cases}$

Suppose $\boldsymbol{w} = [0, -0.6, 0.6]$



Loss Function

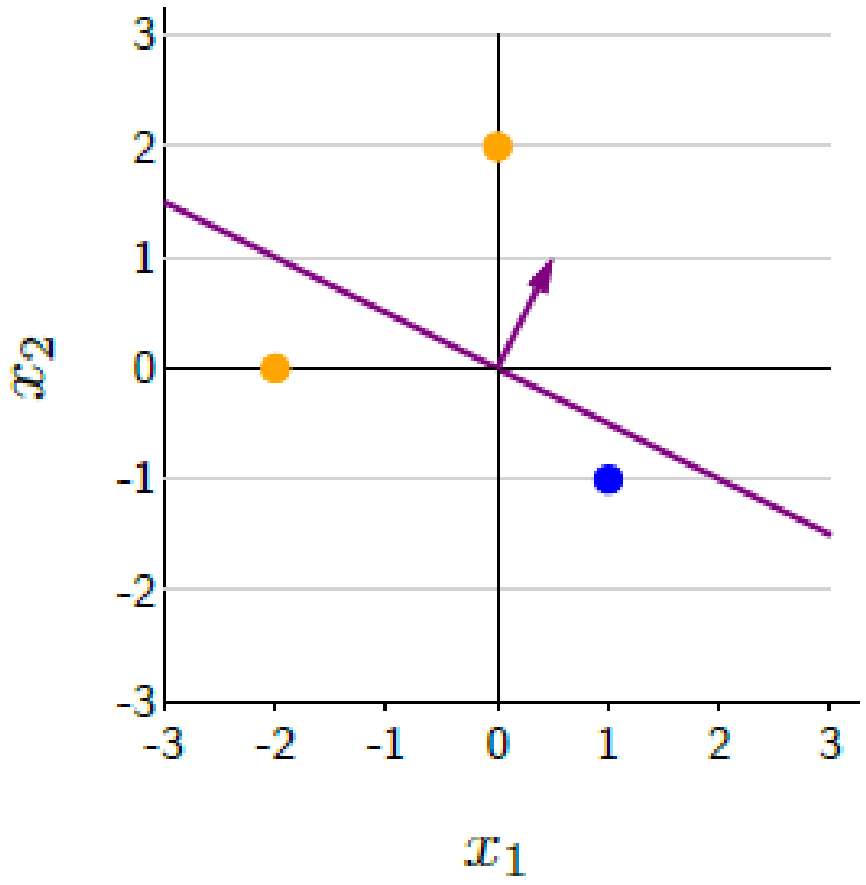
$$h(\boldsymbol{x}) = sign(\boldsymbol{w}^T \phi(\boldsymbol{x}))$$

$$w = [0 \;\; 0.5 \;\; 1]$$

$$\phi(\boldsymbol{x}) = [1 \;\; x_1 \;\; x_2]$$

training data $\mathcal{D}_{\text{train}}$

x_1	x_2	y
0	2	1
-2	0	1
1	-1	-1

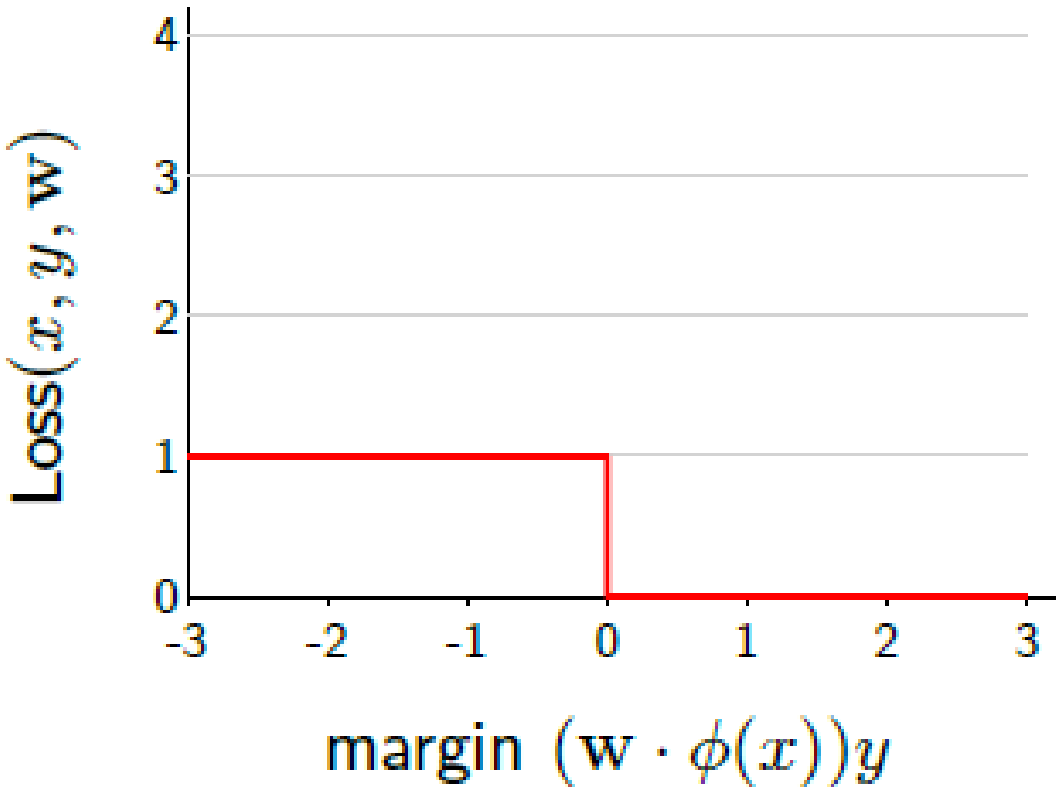


$$Loss_{0-1}(x, y, \boldsymbol{w}) = 1\,[h(\boldsymbol{x}) \neq y]$$

- Total loss is average of all losses over training examples

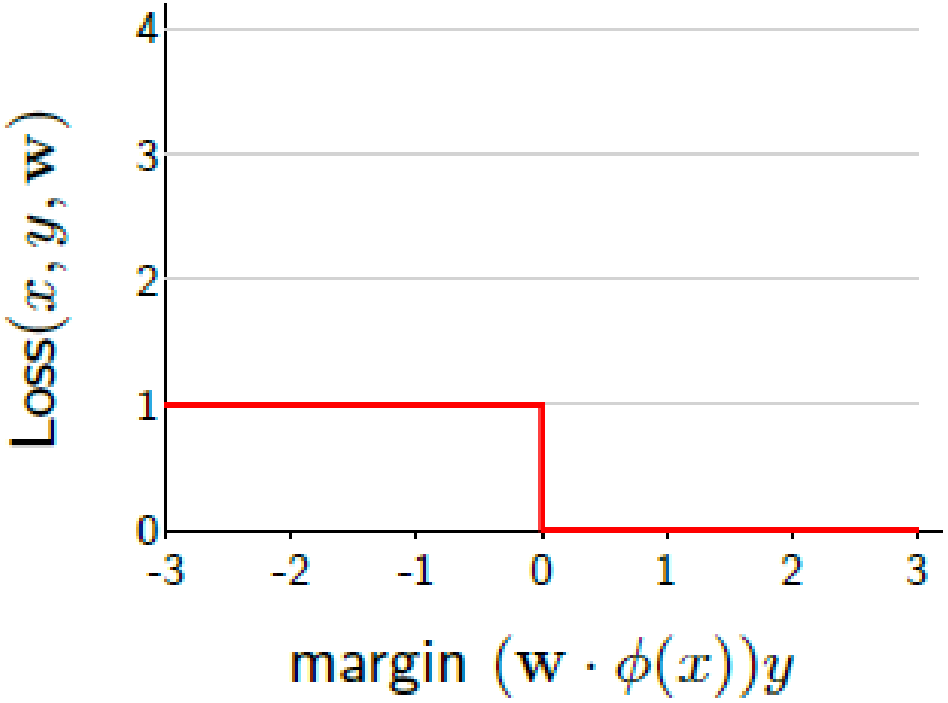
Score and Margin

- Score: $\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x})$ How confident we are in predicting +1
- Margin: $y(\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x}))$ How correct we are.
- Zero-One Loss



Optimization Algorithm

- Goal is to minimise $TrainLoss(\boldsymbol{w})$
- One option is to run gradient descent
- Runs in to problem with zero-one loss. What is the problem?
- Gradient is zero almost everywhere

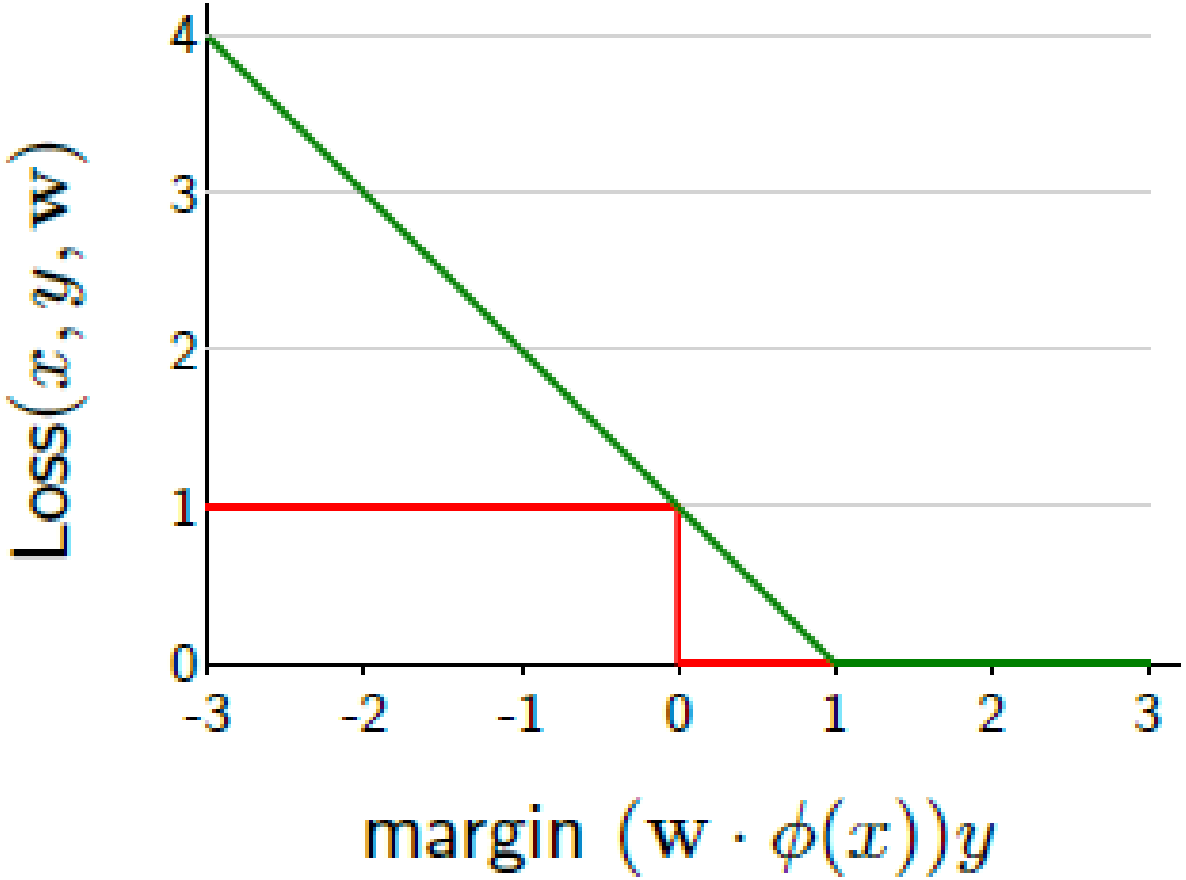


Hinge Loss

- Green line is hinge loss

$$Loss_{hinge}(x, y, \boldsymbol{w}) = \max\{1 - y(\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x})), 0\}$$

- Margin needs to be at least 1 for hinge loss to be zero



Gradient of Hinge Loss

$$Loss_{hinge}(x, y, \boldsymbol{w}) = \max\{1 - y(\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x})), 0\}$$

- What is the gradient of the above function?

$$\nabla Loss_{hinge}(x, y, \boldsymbol{w}) = \begin{cases} -y\phi(\boldsymbol{x}) & if\, y(\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x}) < \mathbf{1} \\ 0 & otherwise \end{cases}$$

Optimization Algorithm with Hinge Loss

- Gradient Descent:

$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} - \alpha \nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w})$$

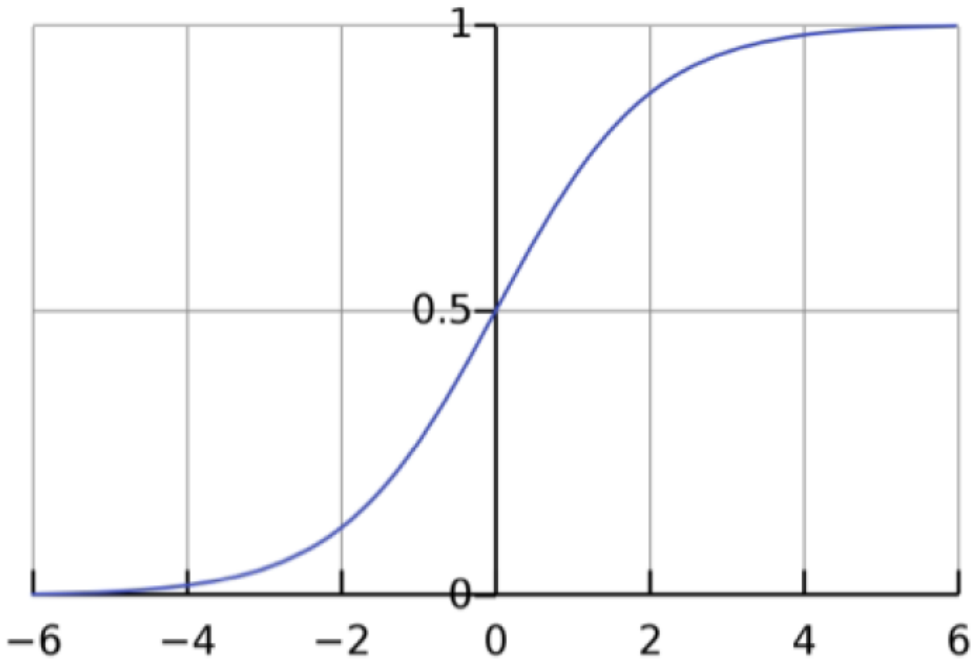
$$\boldsymbol{w}_{new} = \boldsymbol{w}_{old} + \frac{\alpha}{N} \sum_{i=1}^{N} [y_i \phi(\boldsymbol{x}_{\boldsymbol{i}})]$$

Logistic Regression – Another Linear Classifier

- Hypothesis class: $h(\boldsymbol{x}) = g(\boldsymbol{w}^T \phi(\boldsymbol{x}))$

Where $\mathrm{g}(z) = \dfrac{1}{1+e^{-z}}$ (Sigmoid/Logistic Function)

- Output interpreted as probability of positive label.
- $g(z) > 0.5$ means positive label
- $g(z) \leq 0.5$ means negative label



Questions?